

Evaluating the Effects of Radar Characteristics on Coexistent 5G Signals: A Combined Simulation and Experimental Approach

Abstract—The increasing demand for spectrum necessitates the shared use of frequency bands between commercial wireless systems and incumbent radar operations. The 3.5 GHz Citizens Broadband Radio Service (CBRS) band in the United States is a prime example where 5G New Radio (NR) deployments must coexist with federal radar systems. This paper explores the relationship between pulsed sensor interference, particularly its Pulse Repetition Frequency (PRF), and the resulting effects on 5G downlink throughput. We present a comprehensive research approach combining a simulation framework utilizing open-source 5G software stacks and custom radar modeling in MATLAB, validated by over-the-air (OTA) experiments with a commercial UE and Software Defined Radios (SDRs). Our findings demonstrate that 5G and radar coexistence is achievable under certain conditions, and critically, not all radar interference scenarios result in complete blockage of 5G communications. The simulations reveal that for low radar PRFs (below 3 kHz) and under 15% duty cycle, strategic adjustment of 5G symbol-based scheduling can lead to notable improvements in downlink throughput. Furthermore, the results from our real-world OTA testing show good alignment with the simulation outcomes, confirming the viability of our modeling approach. This work identifies specific sensor PRF ranges that are more conducive to cellular coexistence and highlights the potential for preserving 5G performance under high duty cycle conditions by adapting TTI boundaries and symbol allocations.

Index Terms—5G, Radar, Sensor, PRF, PD, Duty Cycle, DL, UL, Scheduler, MAC, MCS, CQI, Spectrum, Co-Existence, Symbol, TTI, Matlab, Model, Harq, srsRAN, open5GS

I. INTRODUCTION

The increasing demand for wireless spectrum is driving new approaches to spectrum utilization, with spectrum sharing emerging as a critical paradigm. In the United States, the establishment of the Citizens Broadband Radio Service (CBRS) in the 3.5 GHz band

(3550–3700 MHz) by the Federal Communications Commission (FCC) [1], [2] exemplifies this trend. This band requires commercial 5G New Radio (NR) deployments to dynamically coexist with incumbent federal radar systems, creating significant interference challenges.

Early efforts to enable coexistence focused on protecting radar operations from cellular emissions. For example, the National Telecommunications and Information Administration (NTIA) conducted link-budget analyses and proposed exclusion zones in its 2010 report [3] to safeguard federal radars in the 3.5 GHz band. However, the FCC’s 2015 Report and Order [4] observed that subsequent NTIA analyses still underestimated the impact of radar-to-cellular interference. The FCC thus encouraged stakeholders to study radar–5G interference further and develop techniques to mitigate harmful radar effects on commercial networks [4].

While considerable research has since investigated 5G–radar coexistence in this band, much of the prior work focuses on the interference characteristics of specific radar types (e.g., particular air, ground, or shipborne radars identified in the NTIA study [3]) and on their fixed duty cycles. The NTIA-defined duty cycles for these incumbent radars range from approximately 0.01% up to 20% [3]. Duty cycle (DC) is defined as the product of a radar’s pulse duration (PD) and pulse repetition frequency (PRF), i.e., $DC = PD \times PRF$ [3]. Because of this relationship, different combinations of PD and PRF can yield similar duty cycle values. Thus, analyses confined to a few specific duty-cycle values may obscure the distinct impacts of varying PRF or PD individually.

There is a need to move beyond studies tied to particular incumbent systems and to understand more generally how fundamental radar parameters influence 5G performance. In this work, we address that need

by treating the radar’s PRF as an independent variable and adopting a generalized approach to radar signal parameters, rather than limiting the study to the fixed duty cycles of known radar systems [3]. This perspective allows us to examine interference scenarios beyond the original NTIA cases and to isolate the effects of PRF on 5G communications.

This paper contributes to filling that research gap by investigating how pulsed radar interference, particularly the pulse repetition frequency, impacts 5G NR throughput. Our approach combines detailed link-level simulations with real over-the-air (OTA) experiments to identify relationships between key radar sensor parameters and 5G performance metrics (with a focus on downlink throughput) [5], [6]. We specifically examine the influence of the radar’s PRF while keeping the pulse duration constant, thereby spanning a range of duty-cycle values comparable to those in the NTIA studies and other prior work [7]. In addition, we explore the potential of leveraging flexible 5G features—such as symbol-level scheduling and dynamic Transmission Time Interval (TTI) allocation [8], [9]—to enhance coexistence under deterministic radar pulse interference.

Through a combined simulation and experimental methodology, our work aims to develop a more generalized understanding of radar interference effects that is not tied to a single incumbent system’s characteristics [10]. By identifying the sensor signal characteristics most detrimental to 5G performance, we evaluate the effectiveness of adaptive, symbol-based scheduling as a mitigation strategy. Our results indicate that strategic, dynamic TTI scheduling in response to radar interference can significantly improve user downlink throughput under certain conditions [11]. Moreover, the insights from this research can help inform future spectrum-sharing policies and guide the design of both radar systems and cellular networks optimized for better coexistence.

The key contributions of this work are as follows:

- **Generalized Coexistence Framework:** We develop an evaluation framework that integrates simulations with over-the-air testing, providing a versatile platform to assess radar–5G coexistence beyond any specific radar waveform or deployment scenario.
- **Isolation of PRF Effects:** By systematically varying the radar pulse repetition frequency (while holding pulse width constant), we isolate PRF

as an independent factor. This approach moves beyond fixed duty-cycle analyses and reveals how different PRFs (and resultant duty cycles) uniquely affect 5G link performance.

- **Symbol-Level Scheduling Mitigation:** We investigate and demonstrate the potential of 5G symbol-level scheduling techniques (adaptive TTI scheduling) as an interference mitigation strategy. The study evaluates how fine-grained scheduling adjustments can sustain throughput under pulsed radar interference.
- **Simulation and OTA Validation:** We validate our findings through both comprehensive simulation and controlled OTA experiments. This dual methodology confirms the accuracy of the simulations and the practical feasibility of our proposed coexistence and interference-mitigation techniques in real-world conditions.

II. RELATED WORK

As 5G networks expand into the midband spectrum, particularly the n78 band (3.3–3.8 GHz) [12]—coexistence with incumbent radar systems has emerged as a critical concern. Research in coexistence of radar and cellular communications falls under two main thrusts: (1) the impact of cellular transmissions on radar performance and (2) the effect of radar interference on cellular network operations.

Cellular Communication Impacts to Radar System: Radar systems operated by U.S. federal agencies in the 3.3–3.8 GHz range are typically categorized as radiolocation or radionavigation services [13]. For instance, the Navy employs shipborne long-range surveillance radars in the 3550–3650 MHz band, the Army uses land-based battlefield weapon-locating radars in the 3100–3400 MHz range, and the Air Force deploys airborne search and surveillance radars across 3350–3600 MHz.

[14] highlighted base station power control as a potential mitigation technique to reduce radar interference. Additionally, LTE transmission power has been shown to correlate directly with increased radar false alarm rates (FAR) [5]. The study notes that many radar systems identified in [13] are pulsed-based, a consideration that motivates our use of FM pulse-based radar models in MATLAB for experimental simulation. *Radar System Interference to Cellular Communications:* Radar interference in OFDM-based networks has been extensively studied, particularly its impact on

5G uplink block collision rates (BCR). Power levels of radar signals directly affect throughput, while radar pulse repetition frequency (PRF) and pulse width influence error patterns and performance degradation [5]. Simulation results consistently show that 5G throughput declines rapidly with increasing radar power.

Both LTE and 5G systems exhibit varying degrees of robustness to radar interference. For example, simulations in [6] show LTE can remain operational even under low SNR conditions caused by radar interference. Experimental work at 1800 MHz confirms a strong correlation between radar power and throughput loss, while pulse width appears to have limited effect [15]. Field studies also reveal that user equipment (UE) is generally more vulnerable to interference than base stations, and TCP-based FTP traffic is more negatively impacted than HTTP traffic [10].

Building on these findings, our study uses a n78 5G NR testbed with TCP traffic and symbol-level scheduling to investigate how varying radar PRF and power levels influence downlink throughput. This enables fine-grained characterization of radar interference effects under realistic deployment conditions.

Additional work has examined radar-induced block error rate (BLER) and symbol error rate (SER) in 5G networks [11], while complementary studies have focused on how 5G signals affect radar performance [16], [17]. These studies, while informative, are beyond the scope of our work.

Some recent efforts have explored the use of 5G signals for passive radar sensing [18]. However, our focus remains on coexistence and interference mitigation rather than enabling sensing applications.

Earlier assessments, from the NITA [3], studied interference from UMTS and WiMAX signals on ground-based and shipborne radar. Subsequent work extended these findings to LTE systems, where a 0.1% duty cycle was used to evaluate radar detection degradation. LTE FDD, LTE TDD, and Gaussian noise were shown to have similar effects on radar detection probability (P_d), with interference-to-noise ratio (I/N) emerging as a key factor [7]. We extend this line of research by systematically varying sensor power levels and PRFs to explore duty cycles beyond 0.1%, offering a more comprehensive view of radar-5G coexistence.

Our work builds on these prior studies by systematically varying the radar pulse repetition frequency (PRF) while keeping the pulse duration fixed. This approach allows us to explore a broad range of radar

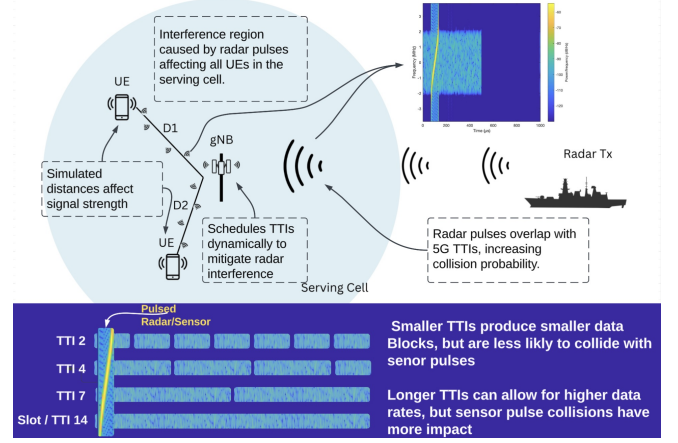


Fig. 1: 5G sensor overlap and Con-Ops, showing TTI Overview and IQ interference and overlap.

duty cycles—from 0% up to 25%—which exceeds the limited duty cycle (e.g., 0.1%) examined in earlier work [7]. By isolating the impact of PRF on interference behavior, we generalize beyond specific radar configurations and provide a more comprehensive characterization of how varying radar activity levels affect 5G system performance.

III. SYSTEM OVERVIEW

To evaluate radar-5G coexistence, we designed a system architecture that enables both simulation-based and OTA-based analysis. Figure 1 illustrates the concept of operations (ConOps), where we model overlapping in-phase and quadrature (IQ) radar interference and analyze its temporal alignment with 5G NR transmissions.

Our framework enables the variation of radar PRF while keeping pulse duration constant. This approach allows us to cover a wide range of duty cycles, from 0% to 25%, extending beyond the specific duty cycles 0.1% explored in prior studies such as [7], and taking a more general approach. By isolating PRF as the key factor contributing to duty cycle, we explore periodic interference more deeply, while also covering a more comprehensive range done by the NTIA[3].

The core idea behind our system is to assess how radar-induced interference aligns with 5G's Transmission Time Interval (TTI), particularly under flexible symbol-based scheduling. By modeling the overlap between radar pulses and 5G symbols at the IQ level, we evaluate the potential for adaptive TTI-based mitigation strategies. The figure shows the radar burst and

TTI alignment, providing intuition on when and how throughput may be impacted depending on PRF values.

This system-level framework enables a deeper understanding of radar interference characteristics and helps identify opportunity spaces where dynamic scheduling can mitigate throughput degradation. Our results show that 5G performance can be preserved even under high PRF conditions by intelligently adapting TTI boundaries and symbol allocations.

IV. EXPERIMENT AND SIMULATION CONFIGURATION

TABLE I: Simulation Parameters

Configuration	Values
gNB Configurations	
Scheduler	Round Robin
Modulation and Coding Scheme (MCS)	64 QAM
Max HARQ	16
Sub-Carrier Spacing	30 KHz
Bandwidth	20 Resource Blocks (10 RBGs)
Slot Pattern	7D1S2U, where S is (8D2G4U)
Slot Pattern Periodicity	5 ms
Scheduler	Round Robin
Sensor Configurations	
Pulse Duration (PD)	50 μ s
Pulse Repetition Frequency Range (PRF)	1 - 5000 (Hz)
Duty Cycle Range ($PRF * PD$)	0-25%
Radar Bandwidth	20 MHz
Radar Offset	Symbol index 2 [1..14]
UE Configurations	
Number of UEs	20
UE Distance from gNB (Simulated)	300m
UE Requested Data Rate (DL/UL)	10Mbps/10Mbps
Max TP Per UE, TTI=2	0.49 (Mbps) 1:20
Max TP Per UE, TTI=4	2.08 (Mbps) 1:20
Max TP Per UE, TTI=7	3.43 (Mbps) 1:20
Max TP Per UE, TTI=14 (Slot Based)	3.78 (Mbps) 1:20

Our MATLAB simulation comprises multiple 10-frame scenarios, each configured according to Table I and incorporating IQ level processing. For every UE scheduling allocation, we generate and overlay radar sensor IQ. Subsequent runs increment a unique sensor characteristic, such as the sensor IQ power level (Figure 2) or the PRF (Figure 4). These simulations are executed for four distinct TTI Scheduling allocations.

The simulation, while including HARQ, incompletely models control plane signaling, applying

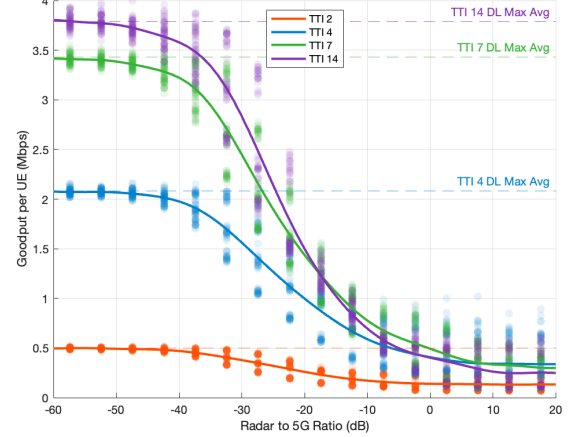


Fig. 2: 20 UEs Simulated throughput against increasing radar sensor power

radar interference primarily to the Physical Downlink Shared Channel (PDSCH). This omission of de-synchronization and control plane message losses from Modulation and Coding Scheme (MCS) changes or Downlink Control Information (DCI) errors may make the simulation appear more interference-robust than a commercial system.

A. Sensor Power level effects

The initial phase of our simulation focused on identifying specific power ratios between the 5G signal and the radar sensor for isolating in later simulations. The effects illustrated in Figure 2 demonstrate phenomena consistent with findings reported in [11] and [5], where the specific power level of the radar signal directly correlates with the throughput experienced by the 5G users. As is evident from the figure, the symbol-based schedulers evaluated exhibit comparable responses to increases in the radar sensor's power level.

To understand the effects of the sensor's Pulse Repetition Frequency (PRF), we chose to isolate the radar power level. We selected this ratio (20dB) for our PRF simulations by selecting the "worst case" scenario, setting it approximately 20 dBm above the 5G signal power while varying the PRF, as illustrated in Figure 3. This approach was based on findings from other studies, which indicate that beyond a certain power level, increasing the radar's power does not result in further jamming of the signal.

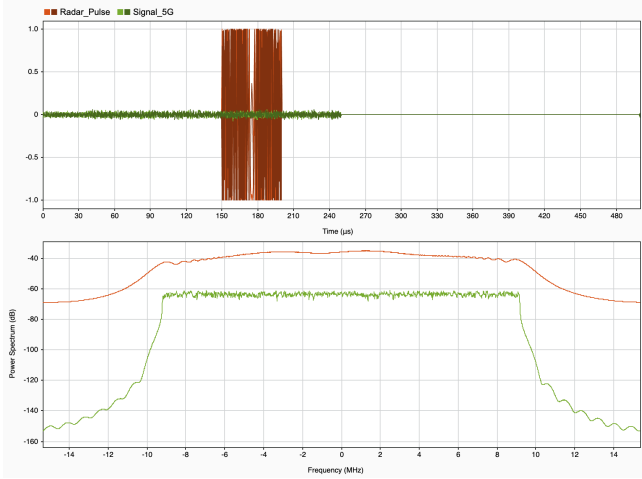


Fig. 3: Example of Radar Pulse (Orange) intersecting with a 5G TTI Burst (Green). The Radar power is 20dB greater than the 5G transmission burst.

B. Sensor PRF Effects

To assess the impact of pulsed sensors on UE performance, we chose goodput as our Key Performance Indicator (KPI). Unlike “throughput” in our MATLAB model, which includes packet retransmissions, goodput specifically reflects successfully delivered data.

Our simulation, configured as outlined in Table I, involves generating sensor IQ at the start of each 10-frame run. This sensor IQ is then mixed with the 5G IQ generated for each user throughout the simulation. Each point in the scatter chart of Figure 4 represents the average goodput of all 20 users for that specific run.

In each simulation iteration, we increment the Pulse Repetition Frequency (PRF), run the 10-frame simulation again, and distill the results into a single average goodput value for the scheduler’s TTI and the sensor’s PRF.

V. OVER THE AIR DESIGN AND DEMONSTRATION

Our OTA experiment (Figure 5) used srsRAN (gNB) and open5GS (5G Core) on a PC with a USRP B200 SDR to emulate a 5G system, testing a Pixel 7 Pro UE. Radar pulses, generated in MATLAB with `phased.CustomFMCWaveform` to create a custom linear FM pulse, were transmitted via a USRP B200 mini. Critical 10 MHz reference clock synchronization ensured precise timing between the gNB and radar SDRs. Both 5G and radar signals passed through a 4:1 RF Combiner and then to the Pixel 7 via a small

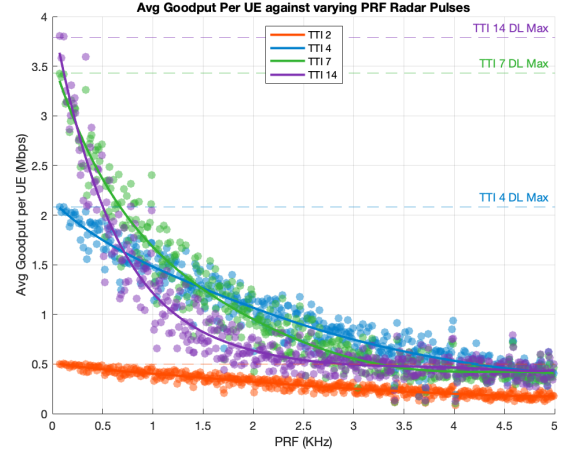


Fig. 4: Simulation Results showing UE throughput under increasing sensor Pulse Repetition Frequencies. Symbol based allocation of TTI 2,4,7 as well as slot (TTI 14)

antenna. We also connected a Lime SDR mini to the combiner to capture combined signal IQ samples using GQRX.

To understand the impacts that pulsed sensors would have we setup the experiment to automate UE throughput testing via `iperf3` and pulse sensors transmission via `matlab`. The procedure (outlined in Figure 6 starts throughput testing for the UE via `iperf`, then begins the pulsed sensors data for 20s. The pulsed sensors are turned off, the throughput returns to max performance, and the logs are collected. We then increment the pulsed sensor prf, and run the experiment again. For each experiment we identify the throughput of the UE while the sensor is present.

Open Source srsRAN at the time of this writing only supports slot based scheduling (TTI = 14), therefore our results show in Figure 7 only compare against TTI = 14 (Purple) as show in Figure 4.

VI. RESULT ASSEMENT

The immediate conclusion from the results posted in Figure 4 is that deterministic pulsed sensor interference does not specifically “jam” UE performance, and cause total disturbance. We can see that each scheduling type is directly impacted by low and higher prf intervals, however all types of scheduling do become saturated to a certain point, and do not cause UE throughput to go zero.

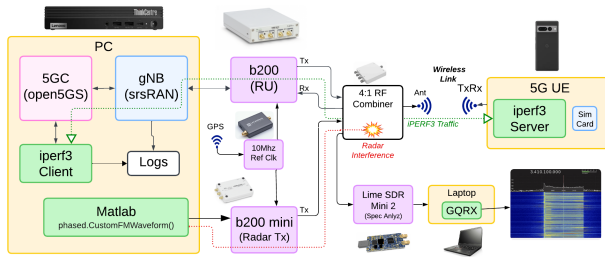


Fig. 5: Over the Air Experiment Setup for generating sensor interference with a commercial UE connected over 5G.

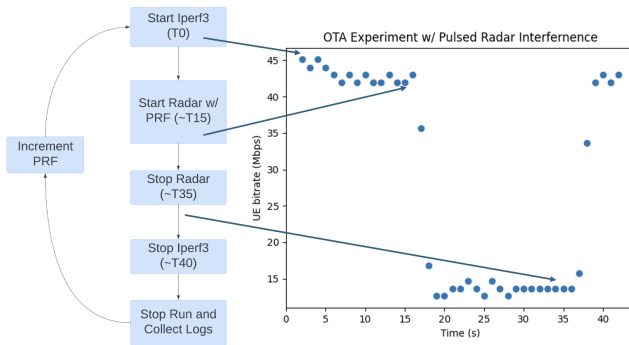


Fig. 6: Automated Experiment procedure showing steps against 500Hz PRF Radar interference

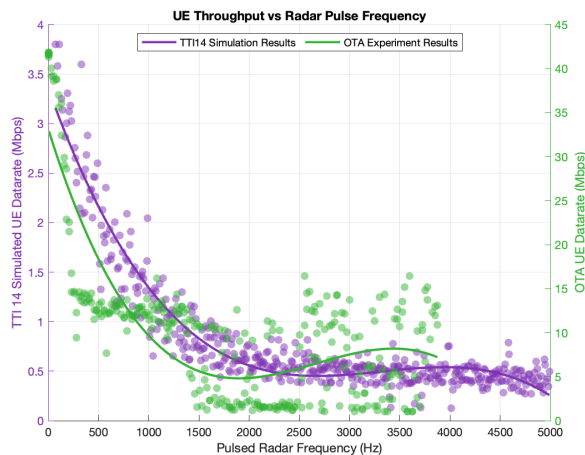


Fig. 7: Experiment Results compared to Simulations

Interestingly the results shown in Figure 4 also show that UE performance can be improved under isolated conditions of both 5G scheduling and sensor PRF. Pulsed sensors with low PRF and duty cycle have high impact for slot based (TTI=14) scheduling. We can see from the trend line TTI 7 (Green) has better performance between 0.5-1.5KHz PRF. For example at 1KHz a TTI of 7 (1.72Mbps) shows a roughly 30% improvement performance per UE rather than TTI 14(1.32Mbps). However still a 65% throughput compared to the slot based max without interference (3.78Mbps)

From the plot TTI 4 (Blue) has the slightly best performance overall from 1.5KHz to nearly 4KHz.

At 2.0KHz we can see that TTI 4 (1.05Mbps) provides a roughly 17% improvement over TTI 7 (0.90Mbps) and a 90% improvement gain per UE if used instead of TTI 14 (0.55Mbps) at this PRF. The smallest symbol allocation, TTI 2 (Orange) shows very little changes over higher PRF. This resilience comes at the tradeoff where the total throughput of each UE is significantly less.

The results in Figure 7 show that real world experiments trend similar to simulation results when comparing slot based scheduling ($TTI = 14$). The throughput loss trends similar between PRFs of 0-500, with a sharp decline in use throughput even at smaller value. When the PRF becomes larger ($> 1.5 KHz$) we can see that the throughput both both curves reaches a relative limit. Another significant validation of the experimental effect is that the UE performance also does not directly go to zero, and cause RLF (Radio Link Failure) under these conditions.

VII. CONCLUSION AND FUTURE WORK

This work presents a systematic investigation into radar-5G coexistence in the CBRS band, focusing on the impact of radar Pulse Repetition Frequency (PRF) and the effectiveness of 5G symbol-level scheduling. Using both simulation and over-the-air (OTA) experimentation, we demonstrate that interference from deterministic pulsed sensors does not completely degrade UE performance, and that certain PRF and scheduling combinations can preserve significant throughput. These findings validate the feasibility of flexible, PRF-aware 5G scheduling as a mitigation strategy.

Our study represents the first stage of a broader, two-stage cognitive-inspired framework for spectrum coexistence. The current stage reflects a “slow-thinking” ap-

proach, emphasizing deliberate and interpretable radar and scheduling parameter design based on deterministic modeling. This provides a robust foundation for a “fast-thinking” second stage, which will incorporate adaptive, real-time response strategies to dynamic interference patterns.

Future work will extend this two-stage cognitive-inspired strategy by deepening both deliberate design and adaptive response capabilities. We plan to incorporate additional metrics such as Radio Link Failure (RLF) and PRACH access failures to capture broader interference effects. The relationship between equivalent duty cycles—produced through different combinations of PRF and pulse duration—will be further explored to generalize interference behavior. We also aim to evaluate how center frequency alignment and bandwidth overlap influence coexistence. Finally, we will investigate machine learning and rule-based methods to infer radar characteristics (e.g., PRF, PD, duty cycle) from gNB-observed KPIs such as BLER and uplink SNR, enabling real-time adaptive scheduling. Additional study will examine other 5G NR structures beyond TTI, including PSS/SSS periodicity, DMRS, CSI-RS, and PUSCH allocation.

REFERENCES

- [1] Federal Communications Commission, “Report and Order and Second Further Notice of Proposed Rulemaking,” Federal Communications Commission, April 2015, adopted: April 17, 2015; Released: April 17, 2015.
- [2] Federal Communications Commission (FCC), “3.5 GHz Band Overview,” 2021, accessed: 2024-11-20. [Online]. Available: <https://www.fcc.gov/wireless/bureau-divisions/mobility-division/35-ghz-band/35-ghz-band-overview>
- [3] U.S. Department of Commerce, National Telecommunications and Information Administration, “An assessment of the near-term viability of accommodating wireless broadband systems in the 1675-1710 mhz, 1755-1780 mhz, 3500-3650 mhz, 4200-4220 mhz and 4380-4400 mhz bands (president’s spectrum plan report),” Tech. Rep., Oct. 2010.
- [4] Federal Communications Commission (FCC), “Wireless Telecommunications Bureau and Office of Engineering and Technology Announce Procedures for Spectrum Access System to Begin Initial Commercial Deployments in the 3.5 GHz Band,” Federal Communications Commission, Tech. Rep. DA 20-110, 2020, accessed: 2024-11-20.
- [5] T. Ranstrom, P. Pietraski, and S. Pattar, “5g pusch channel estimation and decoding subject to high-power pulse radar interference,” in *MILCOM 2022 - 2022 IEEE Military Communications Conference (MILCOM)*, 2022, pp. 437–441.
- [6] M. Ghorbanzadeh, E. Visotsky, P. Moorut, and C. Clancy, “Radar interference into lte base stations in the 3.5 ghz band,” *Physical Communication*, vol. 20, pp. 33–47, 2016.
- [7] F. H. Sanders, J. E. Carroll, G. Sanders, R. L. Sole, R. J. Achatz, and L. S. Cohen, “Emc measurements for spectrum sharing between lte signals and radar receivers,” Institute for Telecommunication Sciences National Telecommunications Information Administration, NTIA Technical Report TR-14-507, July 2014.
- [8] 3GPP, “NR; Physical layer procedures for data,” 3rd Generation Partnership Project, Technical Specification 38.214, Technical Specification Group Radio Access Network.
- [9] —, “NR; Medium Access Control (MAC) protocol specification,” 3rd Generation Partnership Project, Technical Specification 38.321, Technical Specification Group Radio Access Network.
- [10] H.-J. Hong, S. W. Choi, C. Sup Kim, and Y. J. Chong, “Interference measurement between 3.5 ghz 5g system and radar,” in *2018 International Conference on Information and Communication Technology Convergence (ICTC)*, 2018, pp. 1539–1541.
- [11] J. A. DeVault, B. H. Kirk, A. F. Martone, R. M. Narayanan, and K. D. Sherbondy, “A proposed paradigm for evaluating spectrum sharing between a cognitive radar and 4g/5g communications,” in *2022 IEEE Radar Conference (RadarConf22)*, 2022, pp. 1–6.
- [12] *NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 standalone*, 3rd Generation Partnership Project (3GPP) Technical Specification TS 38.101-1, 2019.
- [13] “Federal radar spectrum requirements special publication 00-40,” National Telecommunications and Information Administration (NTIA), Tech. Rep., May 2000.
- [14] S.-S. Raymond, A. Abubakari, and H.-S. Jo, “Coexistence of power-controlled cellular networks with rotating radar,” *IEEE Journal on Selected Areas in Communications*, vol. 34, no. 10, pp. 2605–2616, 2016.
- [15] M. Juskauskas, J. Krivochiza, J. Aleksandravicius, K. Svirkas, B. Dzindzeleta, R. Aleksiejunas, and M. Zilinskas, “Experimental investigation of radar interference into lte system at 1800 mhz frequency band,” in *2013 21st Telecommunications Forum Telfor (TELFOR)*, 2013, pp. 28–30.
- [16] J. A. DeVault, J. A. Kovarskiy, B. H. Kirk, A. F. Martone, R. M. Narayanan, and K. D. Sherbondy, “Lte interference effects on radar performance,” in *2021 IEEE Radar Conference (RadarConf21)*, 2021, pp. 1–6.
- [17] B. Kim, M. kang, H. Kim, and S. Park, “A study of radar interference effects analysis using simulation,” in *2019 International Conference on Information and Communication Technology Convergence (ICTC)*, 2019, pp. 1076–1078.
- [18] A. Ksieżyk, M. Plotka, K. Abratkiewicz, R. Maksymiuk, J. Wszolek, P. Samczyński, and T. P. Zieliński, “Opportunities and limitations in radar sensing based on 5g broadband cellular networks,” *IEEE Aerospace and Electronic Systems Magazine*, vol. 38, no. 9, pp. 4–21, 2023.